

# Detecting Sorbed Hydrocarbons in a Porous Medium Using Proton Nuclear Magnetic Resonance

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An ability to detect residual (sorbed) concentrations of hydrocarbons in a porous medium using nonintrusive geophysical methods is of interest in the remediation of hydrocarbon-contaminated sites. Proton NMR is an noninvasive analytical technique that can be applied to characterize several rock properties in the lab and the field. In this study, we perform a series of laboratory experiments to quantify the effect of sorbed crude oil on the NMR relaxation time ( $T_1$ ) of water-saturated silica gels, both in the presence and absence of iron oxyhydroxide. The experimental data indicate that in the absence of paramagnetic iron impurities,  $T_1$  remains constant in the presence of sorbed hydrocarbons.  $T_1$  increases with increasing hydrocarbon concentration only if the silica surface is coated with a paramagnetic substance such as iron oxyhydroxide. These results suggest that increases in  $T_1$  previously observed for water in natural materials coated with oil are caused by the shielding of paramagnetic surface sites by the sorbed hydrocarbons. The major implication of this study is that proton NMR is a potential means of detecting sorbed or residual hydrocarbons in natural environments.

## Introduction

An ability to detect residual (sorbed) concentrations of hydrocarbons in a porous medium is of interest in the remediation of hydrocarbon-contaminated sites. Noninvasive geophysical techniques, such as proton nuclear magnetic resonance (NMR), might provide an expedient and cost-effective means of detecting these low concentrations of hydrocarbons. In the oil industry, NMR has been used as a borehole logging tool for two decades (1–4), and the recent development of a surface loop instrument permits a variety of near-surface and environmental applications (5–8). NMR has been used, in the lab and in the field, to characterize several rock properties including porosity, pore size distribution, permeability (9–13), and surface wettability (14–22). In this study, we perform a series of laboratory experiments to assess the use of NMR for the detection of residual concentrations of hydrocarbons sorbed to fine silica, with water as the saturating phase.

The theory of proton NMR relaxation is extensively reviewed elsewhere (23, 24). Briefly, a sample is exposed to a large external magnetic field of specific frequency, causing

alignment of the magnetic moments of the protons therein. A radio frequency pulse is then used to excite some of the protons to a higher energy state, and the magnetization ( $M$ ) is monitored over time ( $t$ ) as the system returns (relaxes) to equilibrium. The magnetic relaxation of molecules in a liquid follows an exponential decay

$$M(t) = A \exp(-t/T_{1b}) \quad (1)$$

where the amplitude ( $A$ ) is proportional to the amount of liquid present, and  $T_{1b}$  is the bulk relaxation time of the liquid of interest. In a porous medium, the proximity of the liquid molecules to solid surfaces typically causes the system to relax at a rate that is faster than that of the bulk liquid. For samples with a distribution of pore sizes, the magnetic relaxation curve must be represented by a sum of exponentials, one term for each pore size ( $i$ )

$$M(t) = \sum_i A_i \exp(-t/T_{1i}) \quad (2)$$

where the amplitude  $A_i$  is proportional to the number of molecules in the sample with the relaxation time,  $T_{1i}$ . The observed relaxation times are related to  $T_{1b}$  by

$$\frac{1}{T_{1i}} = \frac{1}{T_{1b}} + \frac{1}{T_{1Si}} = \frac{1}{T_{1b}} + \rho_1 \left( \frac{S}{V} \right)_i \quad (3)$$

where  $T_{1S}^{-1}$  is the contribution of the surface to the relaxation rate of water molecules,  $(S/V)_i$  is the surface area-to-volume ratio of each distinct pore size ( $i$ ), and surface relaxivity ( $\rho_1$ ) indicates the ability of the surface to enhance relaxation.

Several researchers have applied NMR to the detection of hydrocarbons in porous media (14–22, 25–30). They have shown that hydrocarbons (adsorbed or as liquid or gas) can be directly detected by NMR if present in sufficiently high concentration (26–30). Except in the case of Williams and Fung (14), previous studies have shown that the relaxation time of water molecules increases when water-wet surfaces are made oil-wet (16–22). The anomalous results of Williams and Fung (14) have since been shown to be the result of paramagnetic impurities (17), but in general, it has become widely accepted that the changes observed for relaxation time in these studies were the result of wettability differences, even though an alternative mechanism was originally proposed by Brown and Fatt (15). We wish to investigate the change in relaxation times for water when small amounts of sorbed oil are present and to investigate the importance of all possible influences adsorbed oil can have on relaxation time of water in porous media.

There are four dominant mechanisms by which sorbed hydrocarbons may affect NMR response. First, hydrocarbon sorption may change the effective surface area-to-volume ratio of the pores (eq 3). Second, both the hydrocarbon and the water have characteristic values of  $T_{1b}$ . The hydrocarbon may have  $T_{1b}$  which is distinct from that of water and may be present in sufficient quantity to influence the  $T_1$  of the fluid-rock system as a whole. Alternatively, paramagnetic ions may leach from the oil to change  $T_{1b}$  of the water. Third, hydrocarbon sorption may alter the wettability (hydrophobicity) of the mineral surfaces. Most mineral surfaces enhance the relaxation of water, such that  $T_1$  of a water-saturated porous medium is significantly less than  $T_{1b}$  of pure water. If the sorbed hydrocarbons are hydrophobic, they may repel the pore water, limit its interaction with the mineral surfaces, and cause  $T_1$  to increase toward  $T_{1b}$  of water. The final

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mechanism by which hydrocarbon sorption may affect NMR response (15) becomes important where paramagnetic sites populate the surfaces of the porous medium. Paramagnetics are known to strongly influence the relaxation of water in porous media (31–34). Iron(III), often in the form of an oxyhydroxide coating, is the most common paramagnetic substance in natural near-surface, oxidized porous media. An iron oxyhydroxide surface coating greatly increases surface relaxivity and thus decreases  $T_1$  of a water-saturated porous medium (34). However, if the iron coating is covered, or shielded, by sorbed hydrocarbons, its ability to enhance relaxation may be limited. This fourth mechanism is distinct from the third in that the sorbed hydrocarbons are not necessarily hydrophobic; they need not repel the pore water but simply act as a barrier between it and the paramagnetic surface sites (15, 25). In summary, the relationship between hydrocarbon content and  $T_1$  of the system depends on the total quantities of water and hydrocarbon present, their pore-scale locations, their characteristic values of  $T_{1b}$ , and the concentration, location, and form of paramagnetic substances in the system. Each parameter has a different effect on NMR response, and they may all vary simultaneously.

The purpose of this study is to investigate the effect of hydrocarbon sorption on NMR response in model systems that are representative of the oxidized subsurface and to assess the relative importance of the four mechanisms outlined above. We use the method of Daughney (35) to control and quantify the concentration of crude oil hydrocarbons sorbed to water-saturated silica, which we compare to NMR response. We investigate the role of surface iron oxyhydroxide on the relationship between  $T_1$  and sorbed hydrocarbon concentration. In these experiments, concentrations of sorbed hydrocarbon and surface iron are varied simultaneously over a continuum, to better constrain their effect(s) on NMR response. We determine water content, porosity, and surface area of representative samples, which we compare to hydrocarbon content, surface iron concentration, and NMR signal. We also measure contact angle to assess the effect of sorbed hydrocarbons on surface wettability, as a means of evaluating fluid-rock interaction. We evaluate  $T_{1b}$  of the pore fluids and of bulk crude oil, and for all samples, we examine the entire relaxation time distribution in order to distinguish peaks related to relaxation of the hydrocarbon from those related to relaxation of the pore water.

## Experimental Section

**Materials.** Silica gel was selected as an analogue for a naturally occurring unconsolidated geologic material. We chose a model material rather than use naturally occurring materials in order to eliminate any impurities and to ensure complete control over chemical composition and micro- and macroporosity. The grain size distribution (70–230 mesh size = 210–62  $\mu\text{m}$  grain diameter) gives rise to a narrow intergranular pore size distribution (ca. 30–100  $\mu\text{m}$ ). The gel has significant microporosity; the manufacturer's specifications for the material include an average micropore diameter of 6 nm and a surface area exceeding 350  $\text{m}^2/\text{g}$ . Thus, this particular silica gel (obtained from BDH, Toronto, ON) provides a pore structure which approximates that of a natural sand/clay mixture but with a much simpler chemical composition.

The crude oil (Cold Lake bitumen) was supplied by Imperial Oil, Calgary, AB. The bitumen was processed minimally, only enough to remove inorganic particles and water. The asphaltene content of the resulting crude oil exceeds 50%. A different oil would have different sorption characteristics, but the purpose of this study is not to compare adsorption of different oils. All other materials were reagent grade chemicals obtained from Sigma-Aldrich, Oakville, ON.

**Sample Preparation.** Reagent grade silica gel was successively rinsed in 10% HCl, 10% NaOH, and distilled, deionized (DDI) water (18 M $\Omega$ ) and then oven-dried, to remove impurities and homogenize the surfaces. These clean silica gel samples were completely free of paramagnetic impurities with surface relaxivity on the order of  $10^{-3} \mu\text{m/s}$ , 3–4 orders of magnitude lower than relaxivities of most natural geologic materials.

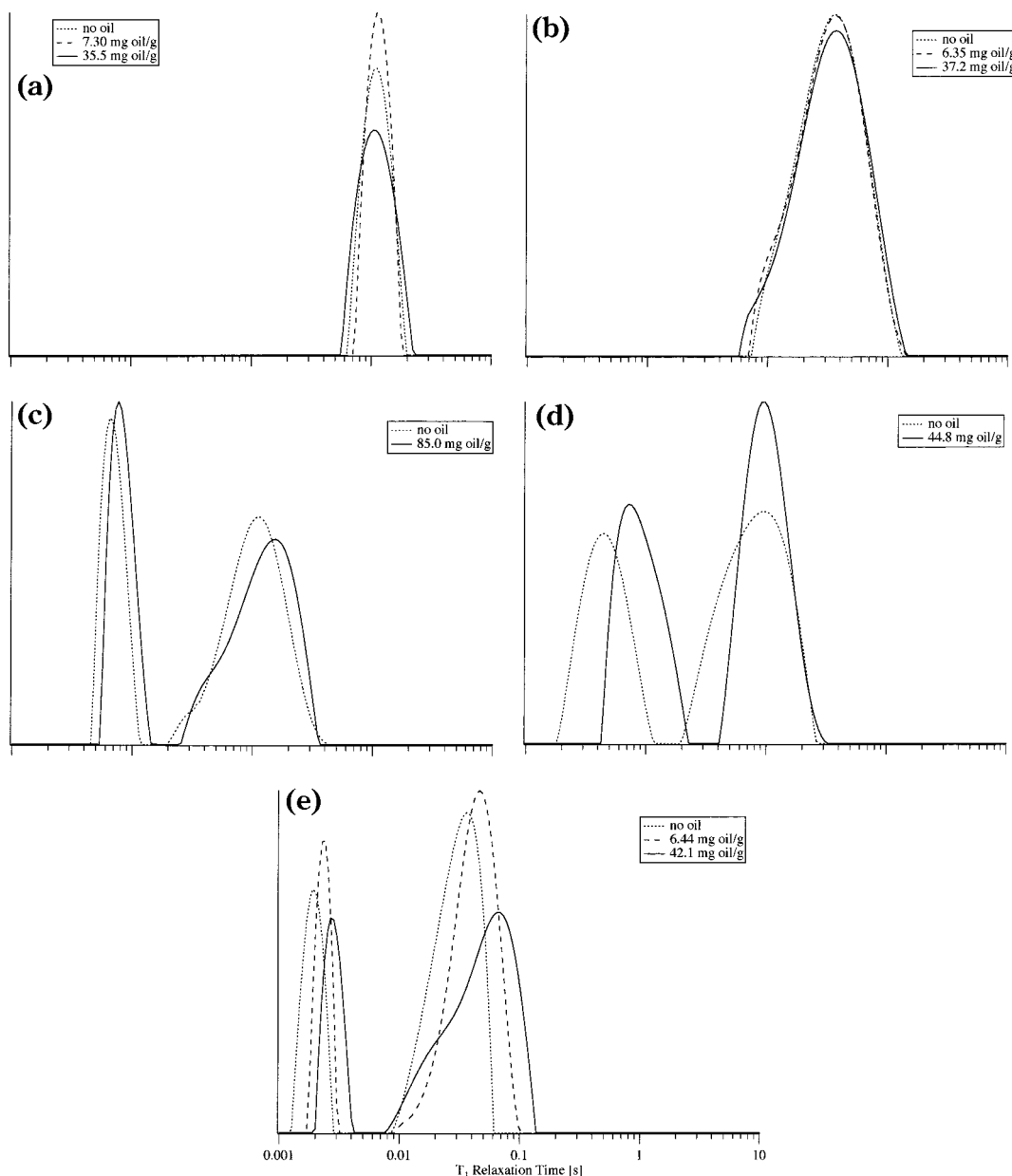
Portions of cleaned silica gel were coated with iron(III) oxyhydroxide using the method of Grantham et al. (36). Mixtures of silica gel and various concentrations of aqueous  $\text{FeSO}_4$  were oxidized with  $\text{H}_2\text{O}_2$ , left standing overnight, rinsed with DDI water, and then oven-dried. The amount of iron on the coated gel was determined by leaching a representative subsample in 10% HCl and analyzing the leachate for dissolved iron by flame atomic absorption spectroscopy. Surface relaxivity of the most iron-rich sample (0.4  $\mu\text{m/s}$ ) approached that of a natural geologic material.

Adsorption of oil by both Fe-coated and uncoated gels was controlled and monitored as described by Daughney (35). The presence of iron has no effect on the amount of oil which adsorbs to the surface. Identical Teflon reaction vessels were used, each containing 2.0 g of gel and 25.0 mL of a 1:1 (v:v) toluene–heptane mixture, with different initial concentrations of oil (5–20 000 mg/L). The suspensions were tumbled for 3 days at 25 °C, after which time 10.0 mL of 0.01 M NaCl (pH 7) were added to each reaction vessel. The reaction vessels were tumbled for an additional 3 days before the organic phase was analyzed for oil by spectrophotometry at 402 nm, allowing the concentration of sorbed oil to be determined by difference.

Portions of aqueous pore fluid and the solid phase, with sorbed hydrocarbons, were transferred from each Teflon reaction vessel to a glass NMR tube and compacted to a known volume with a glass plunger. Thus,  $T_1$  was determined for water saturated samples.  $T_{1b}$  measurements were made on a filtered subsample of the excess aqueous fluid in each reaction vessel.

**NMR Measurements.** Magnetic relaxation data were collected at 25 °C using a 90 MHz proton NMR spectrometer with a 10  $\mu\text{s}$  dead time (Bruker Spectrospin, Mississauga, ON). Data were collected with a modified inversion recovery pulse sequence, in which the free induction decay of a ( $180^\circ - t - 90^\circ$ ) pulse was subtracted from that of a  $90^\circ$  pulse. This pulse sequence is superior to the standard inversion recovery experiment in that it is less sensitive to imperfections in the  $180^\circ$  pulse, and it always gives zero amplitude as  $t$  approaches infinity. For each sample, 30 randomly sequenced, exponentially spaced delay times ( $0.5 \text{ ms} < t < 12 \text{ s}$ ) were stacked 24 times; the wait time between scans was 15 s. Both intensity and standard deviation of the intensity were recorded for each  $t$ ; the signal-to-noise ratio of the data was between 200 and 400.

**NMR Data Inversion.** The magnetic decay (eq 2) was fit to a distribution of 160  $T_1$  values (ranging from 1 ms to 10 s) using several well-established inversion routines (37, 38) and some recently developed (39). The selection of a 160 point partition is not crucial; the finer grid allows for a smoother looking distribution and fewer points do not improve the accuracy of the inversion. Inversion parameters were selected so that each datum was misfit by approximately one standard deviation. Features common to all solutions were regarded as required by the data, not artifacts of the noise or the inversion method. Thus, by comparing the output from the different inversions it was possible to discern which peaks (if any) were artifacts as well as the average peak positions. If the  $T_1$  distributions from different inversions were not similar, that sample was discarded as uninterpretable. The position of each peak in the distributions was calculated as the weighted logarithmic mean  $T_1$ . This was



**FIGURE 1.** Relaxation time distributions as a function of sorbed hydrocarbon concentration, for water-saturated silica gels with surface iron concentrations of (a) 0.00, (b) 0.03, (c) 0.42, (d) 3.37, and (e) 9.50 mg/g. Amplitude is proportional to sample volume, and so to facilitate comparison, all distributions have been scaled to the unit integral.

calculated directly from the amplitudes and  $T_1$  times in the distributions, not from the raw decay curves; log mean  $T_1$  of a peak corresponds to the center of the peak (for a symmetric peak) because the distributions are shown as semilog plots.

**Additional Sample Characterization.** Any observed relationship between sorbed hydrocarbon concentration and  $T_1$  distribution might be due to a change in porosity, surface area, or surface wettability, and so we evaluated these variables for several representative NMR samples. To allow calculation of porosity, sample grain density was determined by helium pycnometer, and sample volume and weight were measured before and after NMR data collection and again after the sample (with or without hydrocarbon) was oven-dried. Surface areas were determined by BET  $N_2$ -adsorption using a Micromeritics ORR 2100D surface area analyzer.

The effect of hydrocarbon sorption on surface wettability was evaluated using a contact angle goniometer (Ramé-Hart Model 100-00-15). It was not possible to measure contact angles on the silica gels directly, due to their microporous

nature. Thus silica glass slides were coated with iron(III) oxyhydroxide and/or hydrocarbon, as described above, and contact angles of 15 water droplets were determined in air. Precipitated iron and sorbed hydrocarbon concentrations could not be determined precisely due to the low surface areas of the slides, and so measured contact angles provide only a qualitative assessment of the wettabilities of the silica gel samples.

## Results and Discussion

Relaxation-time distributions for all interpretable samples were either mono- or bimodal (Figure 1). Before any conclusions can be drawn regarding relaxation of water protons, the  $T_1$  distributions must be examined for evidence of peaks related to the relaxation of the hydrocarbon. The  $T_{1b}$  of the bulk crude oil was found to be 0.3 s, whereas  $T_1$  of the sorbed oil, measured in the absence and presence of water, was 0.15 s and 0.4–0.6 s, respectively; the absence of a distinct peak in the range 0.4–0.6 s with amplitude

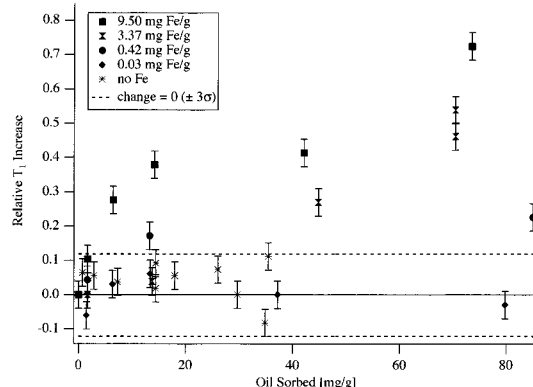
**TABLE 1.  $T_1$  of Water in Uncoated and Iron Oxyhydroxide Coated Silica Gels, with Various Concentrations of Sorbed Hydrocarbons**

| Fe = 0.00<br>mg/g <sup>a</sup> |                    | Fe = 0.03<br>mg/g <sup>a</sup> |                    | Fe = 0.42<br>mg/g <sup>a</sup> |                    | Fe = 3.37<br>mg/g <sup>a</sup> |                    | Fe = 9.50<br>mg/g <sup>a</sup> |                    |
|--------------------------------|--------------------|--------------------------------|--------------------|--------------------------------|--------------------|--------------------------------|--------------------|--------------------------------|--------------------|
| HC <sup>b</sup>                | $T_1$ <sup>c</sup> | HC <sup>b</sup>                | $T_1$ <sup>c</sup> | HC <sup>b</sup>                | $T_1$ <sup>d</sup> | HC <sup>b</sup>                | $T_1$ <sup>d</sup> | HC <sup>b</sup>                | $T_1$ <sup>d</sup> |
| 0.00                           | 1.1                | 0.00                           | 0.31               | 0.00                           | 0.093              | 0.00                           | 0.078              | 0.00                           | 0.029              |
| 0.74                           | 1.2                | 1.48                           | 0.29               | 1.63                           | 0.097              | 1.66                           | 0.078              | 1.65                           | 0.032              |
| 2.87                           | 1.2                | 6.35                           | 0.31               | 6.17                           | NI <sup>e</sup>    | 5.97                           | NI <sup>e</sup>    | 6.44                           | 0.037              |
| 7.30                           | 1.1                | 13.37                          | 0.32               | 13.29                          | 0.11               | 13.76                          | 0.081              | 14.16                          | 0.040              |
| 14.42                          | 1.1                | 37.2                           | 0.31               | 37.3                           | NI <sup>e</sup>    | 44.8                           | 0.099              | 42.1                           | 0.044              |
| 14.42                          | 1.2                | 79.8                           | 0.29               | 85.0                           | 0.11               | 70.4                           | 0.12               | 73.5                           | 0.050              |
| 18.00                          | 1.2                |                                |                    |                                |                    | 70.4                           | 0.11               |                                |                    |
| 26.04                          | 1.2                |                                |                    |                                |                    |                                |                    |                                |                    |
| 29.70                          | 1.1                |                                |                    |                                |                    |                                |                    |                                |                    |
| 34.90                          | 1.0                |                                |                    |                                |                    |                                |                    |                                |                    |
| 35.47                          | 1.2                |                                |                    |                                |                    |                                |                    |                                |                    |
| 74.83                          | NI <sup>e</sup>    |                                |                    |                                |                    |                                |                    |                                |                    |

<sup>a</sup> Concentration of iron as oxyhydroxide precipitated on silica gel (mg/g). <sup>b</sup> Sorbed hydrocarbon concentration (mg/g). <sup>c</sup> Logarithmic mean relaxation time. <sup>d</sup> Log mean  $T_1$  of the slower of two peaks. <sup>e</sup> This relaxation time distribution was not interpretable; i.e., solutions from different inversions of the same data were not similar enough to interpret with any degree of confidence.

proportional to sorbed hydrocarbon concentration in Figure 1a,e, indicates that all peaks in the  $T_1$  distributions are related entirely to the relaxation of water contained in the sample. More importantly, when all water was removed from a sample containing 80 mg oil/g gel, there was too little hydrocarbon sorbed to the surface to be detected by NMR; we found that over 100 mg oil/g gel was required before an oil peak appeared in the  $T_1$  distribution. We can conclude that the  $T_1$  distributions in Figure 1 are related entirely to the relaxation of water.

Bimodal  $T_1$  distributions would be expected for water as a consequence of the bimodal silica gel pore-size distribution if the pores act as isolated regions in terms of their NMR response (eq 3); however, these samples are unconsolidated with micropores and macropores interconnected allowing for the possible diffusion of water molecules between pores of different sizes within the time frame of the relaxation process. If a water molecule in a micropore is relaxed at that micropore's wall, it "sees" only the micropore volume and contributes to the shorter  $T_1$  peak in the bimodal  $T_1$  distribution. However, it is possible for a water molecule located in a micropore to leave the micropore and diffuse across the intergranular space with relaxation occurring on another pore surface. Thus the pore space "seen" by that relaxing proton is the volume of the intergranular macropore rather than the micropore (Figure 1a,b). As a consequence, the slower  $T_1$  peak represents all of the water in the macropores and some of the water in the micropores (Figure 1c–e), so that eq 3 is only an approximation for the relaxation



**FIGURE 2. Relative increase in  $T_1$  as a function of sorbed hydrocarbon concentration, for water-saturated silica gels with different surface iron concentrations. That is, the difference between  $T_1$  of the slower distribution peak in the presence and absence of sorbed hydrocarbons divided by  $T_1$  in the absence of sorbed hydrocarbons.**

behavior of water in unconsolidated samples (40). However, comparisons of  $T_1$  among samples containing different concentrations of adsorbed hydrocarbons are still valid as long as comparisons are made for samples with the same concentration of paramagnetic iron oxyhydroxide.

Similar trends for  $T_1$  were observed for both peaks in the bimodal distributions (Figure 1), so to simplify presentation of the data, the peak positions in the distributions are recorded in Table 1 only for the slower peak (i.e., peak with higher log mean  $T_1$ ). Three trends are apparent in the relaxation-time distributions, and corresponding  $T_1$  values presented in Figure 1 and Table 1. First, in the absence of sorbed hydrocarbons, an increase in surface iron concentration causes  $T_1$  values to decrease and  $T_1$  distributions to shift from mono- to bimodal (compare *no oil* distributions in Figure 1a–e). The  $T_1$  distributions of the saturated untreated and iron-coated gels are in good agreement with previous research (34); the effect of the iron coating on  $T_1$  has been attributed to an increase in surface relaxivity, and the bimodal distributions reflect a partial separation of micro- and macroporosity (34, 40) as discussed above. A second trend is apparent in the absence of an iron oxyhydroxide coating; sorption of hydrocarbons to silica gel does not change relaxation time (see column 1, Table 1), nor does it significantly affect the shape of the  $T_1$  distribution (Figure 1a). This result was unexpected in light of previous work (16–22). As a third trend,  $T_1$  increases with increasing hydrocarbon surface concentration only in the presence of the iron oxyhydroxide coating (compare Figure 1a–e and columns in Table 1); in addition, the relative increase in  $T_1$  is larger for higher concentrations of surface iron (Figure 2). This is the first report of such a trend. However, although a general trend in the data is recognizable, we cannot develop

**TABLE 2. Relative Importance of the Possible Ways Which Adsorbed Hydrocarbon Can Effect  $T_1$  Relaxation in Porous Media**

| mechanism                    | description                                                                                                                | importance in these experiments                                                                        | potential importance in field experiments                                                                                                                                                                  |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $T_{1b}$                     | hydrocarbon could give rise to its own signal<br>hydrocarbon paramagnetics could leach into water, reducing water $T_{1b}$ | no hydrocarbon signal detected<br>no effect – $T_{1b}$ was the same as that of pure water<br>no effect | will be detected if oil present in sufficient amounts<br>if high paramagnetic concentration in oil, then contribution to $T_{1b}$ possible<br>effect small unless large amounts of hydrocarbon are present |
| surface area-to-volume ratio | S/V could decrease if sufficient hydrocarbon covers rough surface                                                          |                                                                                                        |                                                                                                                                                                                                            |
| difference in wettability    | $T_{1S}$ changed by wettability                                                                                            | no effect                                                                                              | no effect                                                                                                                                                                                                  |
| hydrocarbon as a barrier     | $T_{1S}$ changed by changing coupling between water and paramagnetic sites                                                 | important only if paramagnetics present                                                                | most important effect                                                                                                                                                                                      |

a quantitative relationship between  $T_1$ , sorbed hydrocarbon concentration, and surface iron concentration. To explain these trends, we must consider the relative importance of variation in water  $T_{1b}$ , porosity, surface area, and surface wettability.

First, we evaluate the effect of iron oxyhydroxide and adsorbed oil on bulk water  $T_{1b}$ , porosity, and surface area of the samples. In all cases, the bulk liquid, when extracted from the sample, has  $T_{1b} = 2.9 \pm 0.05$  (1 $\sigma$ ) s, characteristic of water in the absence of dissolved paramagnetic ions. The  $N_2$  (BET) physisorption experiments revealed a micropore volume of 0.79 cm<sup>3</sup>/g with a 2–11 nm size distribution confirming the existence of a significant micropore fraction. We calculate the average sample porosity and surface area to be  $0.60 \pm 0.05$  and  $415 \pm 18$  m<sup>2</sup>/g, respectively. There are no significant differences in these averages or standard deviations between the untreated, iron-coated, or oil-coated samples. Thus we conclude that our surface treatments do not affect  $T_{1b}$ , porosity, or surface area and that variations in these parameters are too small to affect the  $T_1$  distributions of the water-saturated samples (see Table 2). Therefore, any differences in  $T_1$  distributions as a result of hydrocarbon adsorption can be attributed to differences in the ability of the surface to enhance relaxation of water protons. Furthermore, bulk water  $T_{1b}$  is too slow to influence  $T_1$  significantly for samples with iron oxyhydroxide. Although the iron-free silica gel samples have very low surface relaxivity ( $\rho = 10^{-3}$   $\mu$ m/s;  $T_1 = 1.1$  s) indicating that  $T_{1b}$  does influence  $T_1$ , surface relaxation ( $T_{1s} = 1.8$  s) still has more of an influence on  $T_1$  than  $T_{1b}$  because of the enormous surface area. Thus, we can be confident that  $T_{1b}$  is not masking a change in  $T_{1s}$  for the case where increasing oil concentration did not change  $T_1$ .

Next, we examine the effect of hydrocarbon sorption on surface wettability. Measurement of the water contact angle on glass slides provides a qualitative means of assessing the wettability of the surfaces. One reason for this is that these samples have slightly less than one monolayer of sorbed oil on the silica surface, thus even if the fraction of the surface coated with oil is hydrophobic, then the surface as a whole is of "mixed" wettability. However, considering the chemical similarity of the surfaces of silica gel and glass, it is reasonable to assume that the results for glass slides will be a good indication of the wettability of the silica gel (if the gel's surface was completely coated with oil). An untreated glass slide, aged in 0.01 M NaCl, displays an average contact angle of 32°. Slides coated with various concentrations of iron oxyhydroxide, then aged in 0.01 M NaCl, have average contact angles of 27–33°. Although the surface iron concentrations were not determined, the slides are reddish after treatment. Immersion of these uncoated and iron-coated slides in toluene–heptane (50:50 by volume), prior to aging in 0.01 M NaCl, does not alter the average contact angle. In contrast, when slides are immersed in a toluene–heptane solution containing various amounts of dissolved oil, then aged in 0.01 M NaCl, the average contact angle increases to between 60 and 75°. Again, sorbed hydrocarbon concentrations were not determined, but brownish staining is visible, and the increase in contact angle is proportional to the initial oil concentration. The change in contact angle upon hydrocarbon adsorption is of similar magnitude for both uncoated and iron-coated slides. Based on these observations, we conclude that the hydrocarbon coatings significantly alter the wettability of the silica gels, whereas the iron coatings do not alter wettability.

Based on the above discussion we conclude that, of all the possible ways in which adsorbed oil could potentially effect NMR relaxation of water protons (see Table 2), only the last is important for our experiments. We have shown that adsorbed oil did not influence  $T_{1b}$  or surface area-to-

volume ratio. In addition, if we accept that silica gel is more hydrophobic with adsorbed hydrocarbon than without, then we have shown that a change in surface wettability alone is not sufficient to explain the observed trends in  $T_1$ . Instead, the surface hydrocarbon acts as a physical barrier, distancing the water protons from the surface paramagnetic sites; as the amount of sorbed oil increased, the fraction of paramagnetic sites shielded by oil increased. A thin layer of sorbed oil significantly alters  $T_1$  through this mechanism, because the effect of a paramagnetic substance on the relaxation of water falls off with distance ( $r$ ) as  $1/r^6$  (24). This mechanism is consistent with the trends shown in Figure 2; not only does  $T_1$  increase with increasing hydrocarbon surface concentration only in the presence of the iron oxyhydroxide coating but also the relative increase in  $T_1$  is larger for higher concentrations of surface iron. The mechanism was proposed by Brown and Fatt in 1956 (15) and later adopted by Devereux (25) to explain his results, but this is the first experimental evidence that physical separations rather than the wetting characteristics of the surface are responsible for sorbed oil's influence on relaxation time of water.

The experimental data presented here indicate that NMR might be useful for the detection of sorbed hydrocarbons in the field. Although  $T_1$  of a water-saturated silica gel only increases with sorbed hydrocarbon concentration in the presence of a surface iron coating, any natural environment will contain sufficient paramagnetic mineral surfaces for this effect to be observable. In the field, spatial heterogeneity in soil properties (e.g. mineralogy, organic matter content) must be taken into account, as this may cause considerable variation in surface relaxivity, porosity, and surface area-to-volume ratio of the pore space, all of which can cause variation in  $T_1$ . Detection of sorbed hydrocarbons should therefore proceed by comparison of relative values of  $T_1$  of contaminated and uncontaminated samples or by repeat measurements over a period of time at the same site.

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## Literature Cited

- (1) Jackson, J. A. *Log Analyst* **1984**, 25, 16–30.
- (2) NMR Bibliography, Society of Professional Well Log Analysts. <http://www.spwla.org/biblio/nmrbib.htm> (current to 1997 when accessed Sept 1999).
- (3) Hodgkins, M. A.; Howard, J. J. *Bull. Am. Assn. Pet. Geol.* **1999**, 83, 114–127.
- (4) Volokitin, Y.; Looyestijin, W. J.; Slijkerman, W. F. J.; Hofman, J. P. Proceedings of the 1999 International Symposium, Golden, CO, Aug 1–4, 1999; Society of Core Analysts: Dallas, TX, paper 9924.
- (5) Schirov, M.; Legchenko, A.; Creer, G. *Expl. Geophys.* **1991**, 22, 333–338.
- (6) Gev, I.; Goldman, M.; Robinovich, B.; Rabinovich, M.; Issar, A. *J. Appl. Geophys.* **1996**, 34, 277–282.
- (7) Legchenko, A.; Beauce, A.; Guillen, A.; Valla, P.; Bernard, J. *Eur. J. Environ. Eng. Geophys.* **1997**, 2, 173–190.
- (8) Weichman, P. B.; Lavelly, E. M.; Ritzwoller, M. H. *Phys. Rev. Lett.* **1999**, 82, 4102–4105.
- (9) Schmidt, E. J.; Velasco, K. K.; Nur, A. M. *J. Appl. Phys.* **1986**, 59, 2788–2797.
- (10) Watson, A. T.; Chang, C. T. P. *Prog. Nucl. Magn. Reson. Spectrosc.* **1997**, 31, 343–386.

- (11) Hinedi, Z. R.; Chang, A. C.; Anderson, M. A.; Borchardt, D. B. *Water Resour. Res.* **1997**, *33*, 2697–2704.
- (12) Davies, S.; Kalam, M. Z.; Packer, J. K.; Zelaya, F. O. *J. Appl. Phys.* **1990**, *67*, 3171–3176.
- (13) Dunn, K. J.; LaTorraca, G. A.; Bergman, D. J. *Geophysics* **1999**, *64*, 470–478.
- (14) Williams, C. E.; Fung, B. M. *J. Magn. Reson.* **1982**, *50*, 71–80.
- (15) Brown, R. J. S.; Fatt, I. *Trans. Am. Inst. Min. Metall. Pet. Eng.* **1956**, *207*, 262–264.
- (16) Kumar, J.; Fatt, I.; Saraf, D. N. *J. Appl. Phys.* **1969**, *40*, 4165–4171.
- (17) Hsu, W. F. Ph.D. Dissertation, Texas A & M University, College Station, TX, May 1994.
- (18) Borgia, G. C.; Fantazzini, P.; Mesini, E. *J. Appl. Phys.* **1991**, *70*, 7623–7625.
- (19) Howard, J. J. *Magn. Reson. Imag.* **1994**, *12*, 197–200.
- (20) Howard, J. J. *Magn. Reson. Imag.* **1998**, *16*, 529–533.
- (21) Kanters, W. A.; Knight, R. J.; MacKay, A. L. *J. Environ. Eng. Geophys.* **1998**, *3*, 197–202.
- (22) Zhang, Q.; Huang, C.; Hirasaki, G. Proceedings of the 1999 International Symposium, Golden, CO, Aug 1–4, 1999; Society of Core Analysts: Dallas, TX, paper 9921.
- (23) Kenyon, W. E. *Log Analyst* **1997**, *38*, 21–43.
- (24) Kleinberg, R. L.; Kenyon, W. E.; Mitra, P. P. *J. Magn. Reson., Ser. A* **1994**, *108*, 206–214.
- (25) Devereux, O. W. *Nature* **1967**, *215*, 614–615.
- (26) Brown, R. J. S. *Nature* **1961**, *189*, 387–388.
- (27) Hedberg, S. A.; Knight, R. J.; MacKay, A. L.; Whittall, K. P. *Water Resour. Res.* **1993**, *29*, 1163–1170.
- (28) Horkowitz, J. P.; Hartman, D. E.; Clerke, E. A.; Vinegar, H. J.; Coates, G. R. *Log Analyst* **1997**, *38*, 73–83.
- (29) Hari, P.; Chang, C. T. P.; Kulkarni, R.; Lien, J. R.; Watson, A. T. *Magn. Reson. Imag.* **1998**, *16*, 545–547.
- (30) Reeves, A. D.; Chudek, J. A. *Water Sci. Technol.* **1998**, *38* (11), 187–192.
- (31) Kenyon, W. E.; Kolleeny, J. A. *J. Colloid Interface Sci.* **1995**, *170*, 502–514.
- (32) Foley, I.; Farooqui, S. A.; Kleinberg, R. L. *J. Magn. Reson., Ser. A* **1996**, *123*, 95–104.
- (33) Roose, P.; Van Craen, J.; Andriessens, G.; Eisendrath, H. *J. Magn. Reson., Ser. A* **1996**, *120*, 206–213.
- (34) Bryar, T. R.; Daughney, C. J.; Knight, R. J. *J. Magn. Reson. A* **1999**, in press.
- (35) Daughney, C. J. *Org. Geochem.* **1999**, in press.
- (36) Grantham, M. C.; Dove, P. M.; DiChristina, T. J. *Geochim. Cosmochim. Acta* **1997**, *61*, 4467–4477.
- (37) Whittall, K. P.; Mackay, A. L. *J. Magn. Reson.* **1989**, *84*, 134–152.
- (38) Whittall, K. P.; Bronskill, M. J.; Henkelman, R. M. *J. Magn. Reson.* **1991**, *95*, 221–234.
- (39) Linseisen, F.; Bryar, T. R. Presented at the 40th Experimental NMR Conference, Orlando, FL, Feb 28–Mar 5, 1999; M&T P 197.
- (40) Ramakrishnan, T. S.; Schwartz, L. M.; Fordham, E. J.; Kenyon, W. E.; Wilkinson, D. J. *Log Analyst* **1999**, *40*, 260–270.

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